

AN ALGORITHM BASED ON A SEQUENCE OF LINEAR COMPLEMENTARITY PROBLEMS APPLIED TO A WALRASIAN EQUILIBRIUM MODEL: AN EXAMPLE

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The Walrasian equilibrium problem is cast as a complementarity problem, and its solution is computed by solving a sequence of linear complementarity problems (SLCP). Earlier numerical experiments have demonstrated the computational efficiency of this approach. So far, however, there exist few relevant theoretical results that characterize the performance of this algorithm.

In the context of a simple example of a Walrasian equilibrium model, we study the iterates of the SLCP algorithm. We show that a particular LCP of this process may have no, one or more complementary solutions. Other LCPs may have both homogeneous and complementary solutions. These features complicate the proof of convergence for the general case.

For this particular example, however, we are able to show that Lemke's algorithm computes a solution to an LCP if one exists, and that the iterative process converges globally.

Key words: Walrasian equilibrium, sequence of linear complementarity problems, numerical example.

1. Introduction

Mathiesen formulated in [7] the Walrasian equilibrium model as a complementarity problem, i.e., find a vector z such that

$$F(z) \geq 0, \quad z \geq 0 \quad \text{and} \quad z^T F(z) = 0,$$

where $F: R^l \rightarrow R^l$. He computed the equilibrium by solving a sequence of approximating linear complementarity problems (LCPs). Each LCP was obtained by taking a first order Taylor expansion, and solved by using Lemke's almost complementary pivoting algorithm.¹

There is strong evidence that this approach is computationally efficient.² More surprisingly, the algorithm appears to be robust as well.³ This is not to say that the algorithm solves globally any Walrasian model with continuously differentiable

¹ Observe that other linearizations are available, resulting in different structures for the LCPs. There also exists alternative algorithms to solve the LCP.

² Applied to a 2-region multiperiod problem, this approach dramatically reduced the CPU-time compared to a fixed point method and a quasi-Newton method, see [11, Figure 5.1], or [12].

³ The experiments with two problems from [15, pp. 109-119] suggested global convergence [7]. (See also [9].)

functions. In fact, Scarf's unstable exchange equilibrium model [14] provides for some parameter values and some starting points, counterexamples to such a proposition.

So far, rather few theoretical results concerning convergence have been obtained. A proof for convergence would have to include both the assertion that the iterates are well defined, i.e., that Lemke's algorithm computes a solution to the LCP, and that a sequence of iterates converges to an equilibrium.

The structure of a typical linear approximation where $F(z)$ is replaced by $q + Mz$ complicates a general proof. It is shown that the matrix M may not conform with any of the types for which Lemke's algorithm is known to compute a solution. Being the gradient of the map F , this matrix does with one exception neither satisfy assumptions that usually are made for proving (local) convergence.

In order to shed some light on the iterative process, we have resorted to studying a specific example. We are able to show that Lemke's algorithm computes a solution if there is one, and that a sequence of iterates converges globally.

The outline of this paper is as follows. In the next section we formulate the Walrasian equilibrium as a complementarity problem and point out some implications of its characteristics for the construction of the approximating LCPs. In Section 3 we present our particular example. Then in Section 4 we establish the LCP and discuss its mathematical structure and what can and cannot be said about solving this problem by Lemke's algorithm. In Section 5 we study the iterates of the process and their convergence. We conclude in Section 6 with some suggestions for further analysis of the proposed algorithm.

2. The Walrasian equilibrium formulated as a complementarity problem

The general equilibrium problem is characterized by the economy's initial endowments, its production technology and the preferences and behavior⁴ of its consumers and procedures.

For $i = 1, \dots, m$, and $j = 1, \dots, n$ let

$p = (p_i)$ denote the vector of prices,

$b = (b_i)$ denote the vector of endowments,

$d(p) = (d_i(p))$ denote the household demand functions, which we assume to be point-to-point and continuously differentiable,

$y = (y_j)$ denote the vector of activity levels and

$A = (a_{ij})$ denote the technology matrix of input-output coefficients consistent with unit production, where $a_{ij} > 0$ ($a_{ij} < 0$) denote an output (input).

We assume that the $m \times n$ technology matrix A does not admit output without input, that is, $\{x \mid x = Ay \geq 0, y \geq 0\} = \{0\}$. Furthermore, we assume that endowments b are finite.

We shall use the following characterization of a general equilibrium (cf. [14, Definition 5.1.3]):

⁴ We assume that all agents take prices as given and consider a competitive equilibrium.

A price vector p^* and a vector of activity levels y^* constitute a general equilibrium if:

$$\text{No activity earns a positive profit;} \quad -A^T p^* \geq 0. \quad (1a)$$

$$\text{No commodity is in excess demand;} \quad b + Ay^* - d(p^*) \geq 0. \quad (1b)$$

$$\text{No prices or activity levels are negative;} \quad p^* \geq 0, \quad y^* \geq 0. \quad (1c)$$

$$\text{An activity earning a deficit is not run, and an operated activity runs at a balance;} \quad (-A^T p^*)^T y^* = 0. \quad (1d)$$

$$\text{A commodity in excess supply has zero price, and a positive price implies market clearance;} \quad p^{*T}(b + Ay^* - d(p^*)) = 0. \quad (1e)$$

In the general equilibrium all prices are determined simultaneously and no single price will be exogenously given. Demand $d_i(p)$, $i = 1, \dots, m$, are functions of all prices in the economy. We assume these functions are specified consistent with individual household utility maximization. That is, $d_i(p) = \sum_j x_j^i$, where x_j^i is the j -th household's utility maximizing demand for commodity i . When the demands satisfy each individual household's budget and there is nonsatiation, then

$$p^T d(p) = p^T b. \quad (WL)$$

This is Walras Law and characterizes the (general) Walrasian equilibrium.⁵ In this case, demands are homogeneous of degree zero. Thus if p^* is an equilibrium price vector, so is λp^* for all $\lambda > 0$.

The stated assumptions on demands have several implications for our construction of approximating LCPs and iterates of the process. First, the homogeneity implies that (1) only determines relative prices. Hence we are free to normalize price-iterates of the process, e.g. price-vectors p , $p \in S = \{p \mid \sum p_i = 1, p_i \geq 0\}$. Nonsatiation and the budget constraint further imply that demand for commodity i is undefined for $p_i = 0$ (at least for some i). Thus the solution to an LCP where $p_i = 0$ can occur may be invalid as an iterate, and we shall have to distinguish between LCP solutions and iterates.

Second, it follows from Euler's equation for homogeneous functions that

$$[\nabla d_i(p)]p = 0, \quad i = 1, \dots, m. \quad (EH)$$

The first order Taylor expansion of $d_i(p)$ at a point $\bar{p}$ is then

$$\begin{aligned} Ld_i(p|\bar{p}) &\equiv d_i(\bar{p}) + [\nabla d_i(\bar{p})](p - \bar{p}) \\ &= d_i(\bar{p}) + [\nabla d_i(\bar{p})]p \quad i = 1, \dots, m. \end{aligned}$$

Furthermore (EH) implies that the Jacobian matrix of demands, $D \equiv \nabla d(\cdot)$, is singular, and hence that the full set of linearized conditions of (1.1)-(1.2) cannot

⁵ Observe that (WL) is a characteristic of the demand functions and holds for all p . It should be distinguished from the similar-looking sum of equilibrium conditions (1d) and (1e), which holds at equilibrium for all demand functions.

compute the equilibrium.⁶ The impasse is traditionally avoided in either of two ways. One can add a normalization constraint, e.g., $\sum_i p_i = 1$, or one can stipulate a numéraire price.

We have chosen the latter approach. Let commodity m , say, be numéraire, that is, stipulate $p_m = \bar{p}_m$. The LCP of the iterative process will then be:

$$\begin{aligned} & \text{find vectors } y \text{ and } \pi \text{ such that} \\ & s - B^T \pi \geq 0, \quad y \geq 0, \quad y^T (s - B^T \pi) = 0, \\ & h + By - G\pi \geq 0, \quad \pi \geq 0, \quad \pi^T (h + By - G\pi) = 0, \end{aligned} \quad (2)$$

where $\pi = (p_1, \dots, p_{m-1})^T$, the matrix B is A without its m th row, G is D without its m th row and column, s is an n -vector with elements $s_j = -a_{mj}\bar{p}_m$, $j = 1, \dots, n$, and h is an $(m-1)$ -vector with elements $h_i = b_i - d_i(\bar{p}) - d_{im}\bar{p}_m$, $i = 1, \dots, m-1$, where d_{im} is an element of the last column of D . Hence (2) is obtained from a linearization of (1) by fixing p_m and dropping the m th commodity balance. (WL) implies that when $m-1$ markets are cleared, the remaining market will also clear.⁷ As there are m commodities, there are m candidates for numéraire.

Observation 1. For any $\bar{p} \in \bar{S} = \{p \mid \sum_i p_i = 1, p_i > 0\}$ there are m alternative LCPs from which a subsequent iterate may possibly be computed.

Let $\text{LCP}(\bar{p})$ denote the LCP obtained by a first order Taylor-expansion at prices $\bar{p}$ and with commodity i as a numéraire.

⁶ Consider the linearization of (1a)-(1b) at $p^* \neq 0$

$$u^* = -A^T p^*, \quad v^* = b - d + A y^* + D p^*, \quad (L)$$

where $d \equiv d(p^*)$ and $D \equiv \nabla d(p^*)$. Partition y^* into subvectors with basic and nonbasic variables and partition the matrix A accordingly. Let B and N denote basic and nonbasic respectively. Rearrange (L) to obtain

$$\begin{pmatrix} I & 0 & A_N^T \\ 0 & 0 & A_B^T \\ 0 & -A_B & -D \end{pmatrix} \begin{pmatrix} u_N^* \\ y_B^* \\ p^* \end{pmatrix} + \begin{pmatrix} y_N^* \\ u_B^* \\ v^* \end{pmatrix} = Mx + w = \begin{pmatrix} 0 \\ 0 \\ b - d \end{pmatrix}. \quad (L')$$

M will be singular if the submatrix

$$Q = \begin{pmatrix} 0 & A_B^T \\ -A_B & -D \end{pmatrix}$$

is singular. From (L'), $A_B^T p^* = 0$ and by (EH), $\nabla d(p^*) p^* = 0$, and hence

$$\begin{pmatrix} A_B^T \\ -D \end{pmatrix} p^* = 0.$$

That is, a nonzero linear combination of vectors of Q is the zero vector. Hence Q and M are singular.
⁷ Existence of an equilibrium and its uniqueness is not the issue here. See [5] for an elegant exposition that is phrased in the same notation.

3. The example

Consider a case with three commodities (one produced commodity and two resources), one profit maximizing producer and one utility maximizing household, which both are price takers. In particular, let the technology matrix A and the initial endowments vector b be given by

$$A = [1 \ 1 \ -1]^T \text{ and } b = (0, b_2, b_3)^T, \text{ with } b_2 > 0, b_3 > 0.$$

Let the household demand functions (as e.g. derived from a Cobb-Douglas utility function) be

$$d_i(p_1, p_2, p_3) = \frac{a_i(b_2 p_2 + b_3 p_3)}{p_i} = \frac{a_i H}{p_i}, \quad i = 1, 2, 3,$$

where $H \equiv b_2 p_2 + b_3 p_3$ denotes income. We observe that these demand functions are well defined on the interior of the price simplex, $\bar{S} = \{p \mid \sum_i p_i = 1, p_i > 0\}$. Finally, let the budget shares of household demand be $a = (a_1, a_2, a_3) = (\alpha, 1 - \alpha, 0)$, with $0 < \alpha < 1$. We have chosen $b_1 = 0$ and $a_3 = 0$ in order to simplify the analysis. The qualitative nature of our results, however, would not be changed by $b_1 > 0$ and $a_3 > 0$. Posed as a complementarity problem, this Walrasian equilibrium problem is:

Find an activity level y and prices p_1, p_2, p_3 such that

$$\begin{aligned} w_0 &= -p_1 + p_2 + p_3 \geq 0, \\ w_1 &= y - \alpha(b_2 p_2 + b_3 p_3)/p_1 \geq 0, \\ w_2 &= b_2 - y - (1 - \alpha)(b_2 p_2 + b_3 p_3)/p_2 \geq 0, \\ w_3 &= b_3 - y \geq 0, \\ y, p_1, p_2, p_3 &\geq 0, \\ w_0 y + w_1 p_1 + w_2 p_2 + w_3 p_3 &= 0. \end{aligned} \quad (3)$$

Depending on the parameter values, (3) has two structurally different equilibria. Let $R \equiv (b_2 - b_3)/b_3$ and $S \equiv (1 - \alpha)/\alpha$. If $R > S$, then $y^* = b_3$ and $p^* = (R/S, 1, R/S - 1)^T$, and if $S > R$, then $y^* = \alpha b_2$ and $p^* = (\frac{1}{2}, \frac{1}{2}, 0)^T$. In particular, when $b = (0, 5, 3)^T$ and $\alpha = 0.9$, $y^* = 3$ and $p^* = (6, 1, 5)^T$. Observe that for all scalars $\lambda > 0$, λp^* are equilibrium prices.

The SLCP algorithm is to solve the nonlinear problem (3) by solving a sequence of approximating linear complementarity problems. The first order Taylor expansions of the nonlinear (demand-) functions, $d_i(p)$, at a point $\bar{p} = (\bar{p}_1, \bar{p}_2, \bar{p}_3)^T$ are

$$\begin{aligned} Ld_i(p|\bar{p}) &= \begin{cases} \alpha[H/\bar{p}_1 - (H/\bar{p}_1^2)\bar{p}_1 + (b_2/\bar{p}_1)\bar{p}_2 + (b_3/\bar{p}_1)\bar{p}_3], & i=1, \\ 0, & i=2, 3, \end{cases} \\ &\equiv \begin{cases} d_1 - d_{11}\bar{p}_1 + d_{12}\bar{p}_2 + d_{13}\bar{p}_3, \\ d_2 - d_{22}\bar{p}_2 + d_{23}\bar{p}_3, \\ 0. \end{cases} \end{aligned}$$

Next, in order to obtain an LCP that possibly could provide an approximate equilibrium, we have to choose a numéraire. The dimension of the problem is thereby reduced by one. There are three commodities and candidates for numéraire. Hence there are three alternative LCPs. As an example, consider $LCP_1(\bar{p})$, which is:

Find variables y, p_2, p_3 such that

$$w_0 = -\bar{p}_1 + p_2 + p_3 \geq 0, \quad (4a)$$

$$w_2 = b_2 - d_2 - y + d_{22}p_2 - d_{23}p_3 \geq 0, \quad (4b)$$

$$w_3 = b_3 - y \geq 0, \quad (4c)$$

$$y, p_2, p_3 \geq 0, \quad (4d)$$

$$w_0 y + w_2 p_2 + w_3 p_3 = 0. \quad (4e)$$

$LCP_1(\bar{p})$ is characterized by:

$$(q^1, M^1) \equiv \left(\begin{pmatrix} -\bar{p}_1 \\ b_2 - d_2 \\ b_3 \end{pmatrix}, \begin{bmatrix} 0 & 1 & 1 \\ -1 & d_{22} & -d_{23} \\ -1 & 0 & 0 \end{bmatrix} \right).$$

Similarly, $LCP_2(\bar{p})$ and $LCP_3(\bar{p})$ are characterized by:

$$(q^2, M^2) \equiv \left(\begin{pmatrix} \bar{p}_2 \\ -d_1 - d_{12}\bar{p}_2 \\ b_3 \end{pmatrix}, \begin{bmatrix} 0 & -1 & 1 \\ 1 & d_{11} & -d_{13} \\ -1 & 0 & 0 \end{bmatrix} \right) \text{ and}$$

$$(q^3, M^3) \equiv \left(\begin{pmatrix} \bar{p}_3 \\ -d_1 - d_{13}\bar{p}_3 \\ b_2 - d_2 - d_{23}\bar{p}_3 \end{pmatrix}, \begin{bmatrix} 0 & -1 & 1 \\ 1 & d_{11} & -d_{12} \\ -1 & 0 & d_{22} \end{bmatrix} \right).$$

Observe that q^i as well as M^i depends on $\bar{p}$. As we shall see below, the structure of M^i alone is insufficient to characterize the performance of the algorithm. Both M^i and q^i matter.

An LCP is said to be *nondegenerate* if any solution $(w_1, w_2, \dots, w_n; x_1, x_2, \dots, x_n)$ of $w = q + Mx$ has at most n zeroes. Observe that except for lines (of measure zero) on the price-simplex $\bar{S}$, $LCP_i(\bar{p})$, $i = 1, 2, 3$, is nondegenerate.

4. Solving the linear complementarity problem

One critical issue for the SLCP algorithm is whether or not a solution can be obtained by solving $LCP_i(\bar{p})$ for some i , $i = 1, 2, 3$. For this purpose we used Lemke's complementarity pivoting algorithm [6].

In the literature there are two types of statements that typically are made regarding an LCP characterized by (q, M) : (1) Lemke's algorithm *solves* the LCP, or (2) Lemke's algorithm *processes* the LCP, that is, computes a solution or shows that none exists. These results obviously depend on (q, M) , and several classes of matrices M have been considered. Let $M \in R^{n \times n}$ and $x \in R^n$. Of importance to us are the following classes [2]:

Class L : $L = L_1 \cap L_2$.

Class L_1 . A square matrix M is in L_1 , if for every x , $0 \neq x \geq 0$ there is an index k such that $x_k > 0$ and $M_k x \geq 0$, where M_k denotes the k th row of M . (M is also called semi-monotone.)

Class L_2 . A square matrix M is in L_2 , if for every x , $0 \neq x \geq 0$, such that $Mx \geq 0$ and $x^T Mx = 0$, there are diagonal matrices $E \geq 0$ and $F \geq 0$ such that $Fx \neq 0$ and $(EM + M^T F)x = 0$.

Eaves [2] has shown that Lemke's algorithm processes the LCP for any q when M is in L .

It is easily checked that M^1 (above) is in L_1 for all $\bar{p} \in \bar{S}$, while M^2 and M^3 are in L_1 for some $\bar{p}$ and not in L_1 for other $\bar{p}$. In fact, M^2 is in L_1 when $\Delta_2 \equiv d_{11} - d_{13} \geq 0$ and M^3 is in L_1 when $\Delta_3 \equiv d_{11} + d_{22} - d_{12} \geq 0$.

Let $\Delta_1 \equiv d_{22} + d_{23}$, which is positive for all $\bar{p} \in \bar{S}$. Observe that $\Delta_i = \det(M^i)$, $i = 1, 2, 3$.

One can also verify that M^1 is in L_2 and hence in L . In fact, vectors $x = \lambda(0, \bar{p}_2/\bar{p}_3, 1)$ for $\lambda > 0$ (that is, $x_2 = (d_{23}/d_{22})x_3$), admit $M^1 x = (x_2 + x_3, 0, 0) \geq 0$ and $x^T M^1 x = x_2(d_{22}x_2 - d_{23}x_3) = 0$. Construct diagonal matrices $E = (e_i) \geq 0$ and $F = (f_i) \geq 0$, where $e_1 > 0$, $f_2 = 0$ and $f_3 = e_1(\bar{p}_2 + \bar{p}_3)/\bar{p}_3$. Then $Fx \neq 0$ and $(EM^1 + M^{1T}F)x = (0, 0, 0)$.

Now consider M^2 and M^3 . One can show that M^2 is in L_2 when $\Delta_2 > 0$, but not in L_2 for $\Delta_2 = 0$. The case for M_3 is similar. When $\Delta_3 > 0$, there is no x , $0 \neq x \geq 0$ that admits $M^3 x \geq 0$ and $x^T M^3 x = 0$. Hence M^3 is in L_2 . When $\Delta_3 = d_{11} + d_{22} - d_{12} = 0$, the vectors $x = \lambda(d_{22}, 1, 1)$, $\lambda > 0$ admit $M^3 x \geq 0$ and $x^T M^3 x = 0$, but there are no matrices $E \geq 0$ and $F \geq 0$ to fulfill the requirements, and M^3 is not in L_2 .

Our conclusion, then, is that Eaves' result tells us that Lemke's algorithm processes $LCP_i(\bar{p})$ for all $\bar{p} \in \bar{S}$ and $LCP_i(\bar{p})$, $i = 2, 3$ for some $\bar{p}$. This result, however, does not guarantee that these LCPs have solutions. Exploiting their structures, that is, both q and M , we shall obtain stronger results.

In order to apply Lemke's algorithm to these LCPs, we add an artificial variable θ and an associated column vector e_i , where $e_i = 1$ if $q_i \leq 0$ and $e_i = 0$ else. When initiated at (the primary ray) $\theta > 0$, $w \geq 0$, $z = 0$, it is well known (see [1]), that Lemke's algorithm terminates in either of two ways: with either a complementary

solution ($\theta = 0$) or a secondary ray ($\theta > 0$). The latter type of termination has the following implications: First, there exists an almost complementary solution (ac-sol) (w^a, z^a, θ^a) that satisfies

$$w^a = q + Mz^a + e\theta^a, \quad z^a, w^a \geq 0, \quad \theta^a > 0, \quad w^{aT}z^a = 0. \quad (5)$$

Next, there exists a homogeneous solution (h-sol) (w^h, z^h, θ^h), with $z^h \neq 0$, that satisfies

$$w^h = Mz^h + e\theta^h, \quad z^h, w^h \geq 0, \quad \theta^h \geq 0, \quad w^{hT}z^h = 0. \quad (6)$$

Finally, the ac-sol and h-sol satisfy $(w^a + tw^h)^T(z^a + tz^h) = 0$ for all $t > 0$, from which it follows that

$$w^{aT}z^a = w^{aT}z^h = w^{hT}z^a = w^{hT}z^h = 0. \quad (7)$$

With this apparatus in hand we can state the following result:

Proposition 1. *Lemke's algorithm computes a solution to $LCP_1(\bar{p})$.*

The proposition is proved by showing that there exists no secondary ray. Hence Lemke's algorithm will terminate with a solution.

Proof. First, observe (by applying (6) to (4.c)) that $w_3^h = -y^h \geq 0$ implies that $y^h = 0$. Next, it follows from (4.c) that w_3^a or y^a has to be positive in an ac-sol. Assume that $w_3^a > 0$. From (7), $w_3^a p_3^h = 0$, hence $p_3^h = 0$. In an h-sol, $w_2^h = -y^h + d_{22}p_2^h - d_{23}p_3^h + e_2\theta^h = d_{22}p_2^h + e_2\theta^h$. Now, $w_2^h p_2^h = 0$ implies that $p_2^h = 0$. But then, $z^h = (y^h, p_2^h, p_3^h)^T = (0, 0, 0)^T$ and this is not a secondary ray.

Alternatively, assume that $y^a > 0$. Then, by applying (7), $0 = w_0^h y^a = (p_2^h + p_3^h + \theta^h)y^a$. Because all variables in the parenthesis are nonnegative, they all have to be zero, that is, $z^h = (y^h, p_2^h, p_3^h)^T = (0, 0, 0)^T$, which again makes us conclude that there is no secondary ray. $\square$

Observe that M^1 allows the nonzero h-sol $z^h = (y^h, p_2^h, p_3^h)^T = (0, \bar{p}_2/\bar{p}_3, 1)^T$, but that q^1 does not allow an ac-sol for which (7) is satisfied. Hence q^1 as well as M^1 is of importance for this proof.

Proposition 1 guarantees that for any price-vector $\bar{p} \in \bar{S}$, Lemke's algorithm applied to $LCP_1(\bar{p})$ computes a solution, and we can obtain a new iterate. The remaining question of convergence of a sequence of such iterates will be discussed in the next section.

Turning now to commodities 2 or 3 as numéraire,⁸ we cannot in general rule out the existence of a secondary ray to the resulting LCP, although such a ray only exists for particular data sets. This will be illustrated by applying Lemke's algorithm

⁸ In order to compute the equilibrium we do not have to consider these numéraires. They do, however, provide valuable insight into the possible events that may occur when applying Lemke's algorithm to such LCPs.

to $LCP_2(\bar{p})$. Posed on the format $Iw - e\theta - Mz = q$, the LCP and the pivoting sequence are shown in Table 1. The columns corresponding to slack variables w are dropped, and the pivot elements are starred.

In (I), d_{11} is the pivot if $\bar{p}_2 \geq (d_1 + d_{12}\bar{p}_2)/d_{11}$ or equivalently $\bar{p}_1 \leq \bar{p}_2/(1 + b_2\bar{p}_2/H)$. If so, the process terminates with a complementary solution, where $(y, p_1, p_3)^T = (0, +, 0)^T$, i.e., p_1 positive. Let CS(1) denote the set of $\bar{p}$'s leading to this complementary solution.

Similarly, in (II) there is another pivot in row 2 if $b_3 \geq d_1 + (d_{12} - d_{11})\bar{p}_2$. This pivot results in the complementary solution $(y, p_1, p_3)^T = (+, +, +)^T$. Let CS(2) denote the set of $\bar{p}$'s leading to this complementary solution.

The final tableau also offers two possibilities. If $\Delta_2 \equiv d_{11} - d_{13} > 0$, we obtain the complementary solution, $(y, p_1, p_3)^T = (+, +, +)^T$, otherwise the process terminates with a secondary ray. Let CS(3) and RAY denote the sets of $\bar{p}$'s leading to the complementary solution and the secondary ray respectively.

These four possible terminating states of the pivoting sequence are illustrated on the price simplex in Figure 1. To describe these regions, consider the following inequalities of the pivoting sequence:

$$\bar{p}_1 \leq \bar{p}_2/(1 + b_2\bar{p}_2/H), \quad (8)$$

$$b_3\bar{p}_1 \geq \alpha[H(1 - \bar{p}_2/\bar{p}_1) + b_2\bar{p}_2], \quad (9)$$

$$b_3\bar{p}_1 \geq (b_2\bar{p}_2 + b_3\bar{p}_3) = H, \quad (10)$$

then

$$CS(1) = \{\bar{p} \in \bar{S} \mid (8) \text{ holds}\},$$

$$CS(2) = \{\bar{p} \in \bar{S} \mid (8) \text{ fails, (9) holds}\},$$

$$RAY = \{\bar{p} \in \bar{S} \mid (8) \text{ and (9) fail, (10) holds}\}$$

$$CS(3) = \{\bar{p} \in \bar{S} \mid (8), (9) \text{ and (10) fail}\}.$$

Table 1
Solving $LCP_2(\bar{p})$ by Lemke's algorithm

	θ	y	p_1	p_3	RHS
(0)	-1*	-1	1	-1	$\bar{p}_2$ $-d_{11} - d_{12}\bar{p}_2$ b_3
(I)	1	1	1*	-1	$\bar{p}_2$ $d_1 + d_{12}\bar{p}_2$ b_3
(II)	1	1	1	-1	$\bar{p}_2$ $d_1 + (d_{12} - d_{11})\bar{p}_2$ b_3
(III)	1	1	1	-1	$\bar{p}_2$ $d_1 + (d_{12} - d_{11})\bar{p}_2 - b_3$ b_3

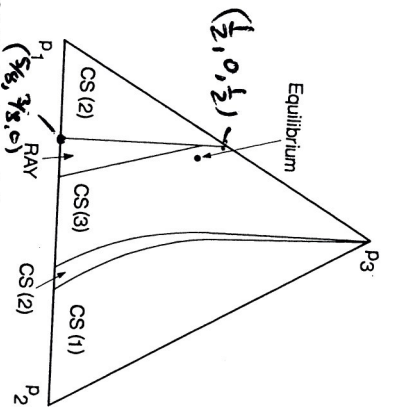


Fig. 1. Terminating states of $LCP_2(\bar{p})$ depending on the point of linearization, $\bar{p}$, on the price simplex.

For the exact demarcation of the regions in Figure 1 the parameter values $b = (0, 5, 3)^T$ and $\alpha = 0.9$ are used.

There are several observations to make from Table 1 and Figure 1. First, there are LCPs that have a nondegenerate secondary ray. When $b = (0, 5, 3)^T$ and $\alpha = 0.9$, some of these LCPs are even evaluated 'close' to the equilibrium, e.g., at $\bar{p} = (0.53, 0.07, 0.40)^T$ while $p^* = (0.5, 0.0833, 0.4167)^T$. Second, the LCPs where Lemke's algorithm terminates with a secondary ray do not have a complementary solution—although they may have feasible, but noncomplementary solutions. Third, M^2 admits homogeneous solutions for all $\bar{p}$ (as did M^1). Thus, there are LCPs for which there exist both a homogeneous solution and a complementary solution.

Fourth, Lemke's algorithm computes a solution to $LCP_2(\bar{p})$ if there is one. Fifth, these LCPs may have more than one solution. If $\bar{p}$ satisfies (9) and (10), i.e., $\bar{p}$ lies in the lower left CS(2)-corner of the simplex, the LCPs have two complementary solutions. One is $(y, p_1, p_3)^T = (+, +, 0)^T$, as computed above by Lemke's algorithm. The other is $(y, p_1, p_3)^T = (+, +, +)^T$, which is in CS(3).⁹

Finally, when $b = (0, 5, 3)^T$, we can compute that $LCP_2(\bar{p})$ has a secondary ray for some $\bar{p}$ for the budget share $\alpha \geq \frac{5}{7}$, while there is no such ray for $\alpha < \frac{5}{7}$.¹⁰

The status of $LCP_3(\bar{p})$ is qualitatively the same as $LCP_2(\bar{p})$. In particular, when $b = (0, 5, 3)^T$, for the budget share α larger than (approximately) 0.63 there is no secondary ray, while $LCP_3(\bar{p})$ has a secondary ray for some $\bar{p}$ for smaller α -values. Note that although some of the above results are based on particular parameter values, Table 1 holds for all $b_2 > 0$, $b_3 > 0$ and $0 < \alpha < 1$. Of course, the sizes and even the existence of some of the regions in Figure 1 will depend on these parameters.

⁹ When $\bar{p}$ satisfies (9) and (10), the solution $(y, p_1, p_3)^T > 0$ cannot be computed by Lemke's algorithm because its index, defined as $\text{sgn}(\det M^2)$, is -1 (see [5]). This is a common property of fixed-point algorithms.

¹⁰ When α is reduced, CS(2) in the lower left corner expands and RAY similarly shrinks. At $\bar{p} = (\frac{5}{6}, \frac{7}{6}, 0)^T$, $\alpha = \frac{5}{7}$ is the maximal value that is consistent with RAY.

In particular RAY will vanish for some parameter values. The qualitative nature of the observations is unchanged, however, and this fact makes us conclude that for the example (3):

Proposition 2. *Lemke's algorithm computes a solution to $LCP_i(\bar{p})$, $i = 1, 2, 3$, if there is one.*

Proposition 3. *For any $\bar{p} \in \bar{S}$, there is at least one $LCP_i(\bar{p})$ (for some i), that has a solution.*

Termination with a secondary ray for $LCP_i(\bar{p})$ means that no new iterate can be computed. A 'recovery' routine would then be to try solve $LCP_j(\bar{p})$ for some $j \neq i$. With this observation, Propositions 2 and 3, taken in conjunction, state that iterates of the SCLP process that switches numéraire, are well defined for the example problem (3).

5. Convergence of the iterative sequence

This chapter concerns the sequence of price iterates $\{p^k\}$. The solution of $LCP_i(\bar{p})$ (if there is one) involves the values p_j^c , $j \neq i$. In addition, we use $\bar{p}_i$ to obtain a complete vector, e.g., for $LCP_2(\bar{p})$, we take $(p_1^c, \bar{p}_2, p_3^c)^T$ to be the price solution. Let $\tilde{p}$ denote the augmented and scaled solution, i.e.,

$$\tilde{p}_j = p_j^c / r, \quad j \neq i, \quad \text{and} \quad \tilde{p}_i = \bar{p}_i / r,$$

where

$$r = \sum_{j \neq i} p_j^c + \bar{p}_i. \quad 11$$

If $\tilde{p} \in \text{domain } d(\cdot)$, then $p^k = \tilde{p}$, that is, we take $\tilde{p}$ as the iterate. $\tilde{p}_j$ may, however, be outside the domain of the demand functions, e.g., $\tilde{p}_j = 0$ for $j = 1, 2$, and if so, cannot serve as an iterate. In such cases we will construct a modified iterate

$$p^k = p^{k-1} + t(\tilde{p} - p^{k-1}), \quad 0 < t < 1.$$

What can be said about the convergence of the sequence $\{p^k\}$ of iterates? As far

¹¹ Scaling of $\tilde{p}$ does not affect $d(\tilde{p})$, because of homogeneity, but it affects coefficients of $\nabla d(\tilde{p})$. (Doubling of prices $\tilde{p}$ doubles the LCP solution p^c but has no effect on relative prices or activity levels.) Our main reason for scaling here, however, is the exposition of the iterative process on the price simplex.

as we know, conventional convergence theory¹² is of little help. There is, however, one relevant result based on the concept of "strong regularity." See [4] and [13]. Applied to our example it guarantees local convergence when M^i is nonsingular at the equilibrium, which we note is the case for $i = 1, 2, 3$.

We will demonstrate global convergence by inspection of the possible iterates of the SLCP process. Consider $LCP_2(\bar{p})$, and the solution obtained for some $\bar{p} \in CS(1)$; viz., $\bar{p} = (1/r)(p_1^e, \bar{p}_2, p_3^e)^T = (1/r)(\bar{p}_1\tau, \bar{p}_2, 0)^T$, where $r \equiv \bar{p}_1\tau + \bar{p}_2$ and $\tau \equiv 1 + 5\bar{p}_2/H$. By choice of pivot, $\bar{p}_1\tau < \bar{p}_2$. Thus, because $\tau > 1$, the scaled solution is located on the right half of the bottom boundary of the simplex in Figure 1. Furthermore, $\bar{p}_1 = (\tau/r)\bar{p}_1 > \bar{p}_1$ when $\tau > r$, which follows because $\tau > 1$ and $1 - \bar{p}_1 \geq \bar{p}_2$ (on the simplex). $\bar{p}_2$, however, may be larger or smaller than $\bar{p}_2$.

If $\bar{p}$ is still in $CS(1)$, the next iteration based on $\bar{p} = \bar{p}$ will yield another solution on the boundary with a higher p_1 value. In fact, for $\bar{p} \in CS(1)$ with $\bar{p}_3 = 0$, the solution will imply $\bar{p}_1 = 2\bar{p}_1/1 + \bar{p}_2 > \bar{p}_1$ because $\bar{p}_1 < \frac{1}{2}$. It is easy to check that this process will eventually reach a point in $CS(2)$ or $CS(3)$, but RAY cannot be reached.

For $\bar{p} \in CS(2)$, the solution is $\bar{p} = (\frac{1}{2}, \frac{1}{2}, 0)^T$, which is in $CS(3)$. For $\bar{p} \in CS(3)$, the solution to $LCP_2(\bar{p})$ satisfies $y^e = b_3 > 0$ and by complementarity $\bar{p}_1 = \bar{p}_2 + \bar{p}_3$, which is a line on the simplex. Successive iterates converge to the equilibrium along this line. In $CS(3)$, the "strong regularity" of M^2 , which here boils down to nonsingularity, guarantees local convergence. Hence the process based on $LCP_2(\bar{p})$ converges to the equilibrium for all $\bar{p} \in \bar{S} \setminus \text{RAY}$.

If $LCP_1(\bar{p})$ or $LCP_3(\bar{p})$ is used, a similar analysis shows convergence to the equilibrium. Hence we have

Proposition 4. *For the example of Section 3, with $b = (0, 5, 3)^T$ and $\alpha = 0.9$, the SLCP process which switches numeraire if necessary converges to the equilibrium for any $\bar{p} \in \bar{S}$.*

(It appears that the proposition could be extended to include all $b_2, b_3 > 0$ and $0 < \alpha < 1$, but that is not checked.)

¹² In [10] a convergence theory for various linear iterative schemes to solve nonlinear complementarity problems is developed. Four approaches to establish the desired convergence are considered; the symmetry approach, the norm-contraction approach, the vector-contraction approach and the monotone approach. The symmetry approach is discarded cause the necessary condition that the gradient matrix $\nabla F(x)$ be symmetric is usually not satisfied in equilibrium models. See (4), i.e., $(LCP_1(\bar{p}))$, where

$$M^1 = \nabla F(x) = \begin{pmatrix} 0 & 1 & 1 \\ -1 & d_{22} & -d_{23} \\ -1 & 0 & 0 \end{pmatrix}.$$

Next, we observe that the gradient matrix in this case is not positive definite as required by the norm-contraction approach, nor is it an H -matrix with positive diagonals (the vector-contraction approach). Furthermore, we note that $\nabla d(p)$, i.e., the Jacobian matrix of demand functions, is not positive definite. Hence Haaves' local convergence results do not apply. (See [3].) Finally, $F(x)$ is concave and not convex (the monotone approach).

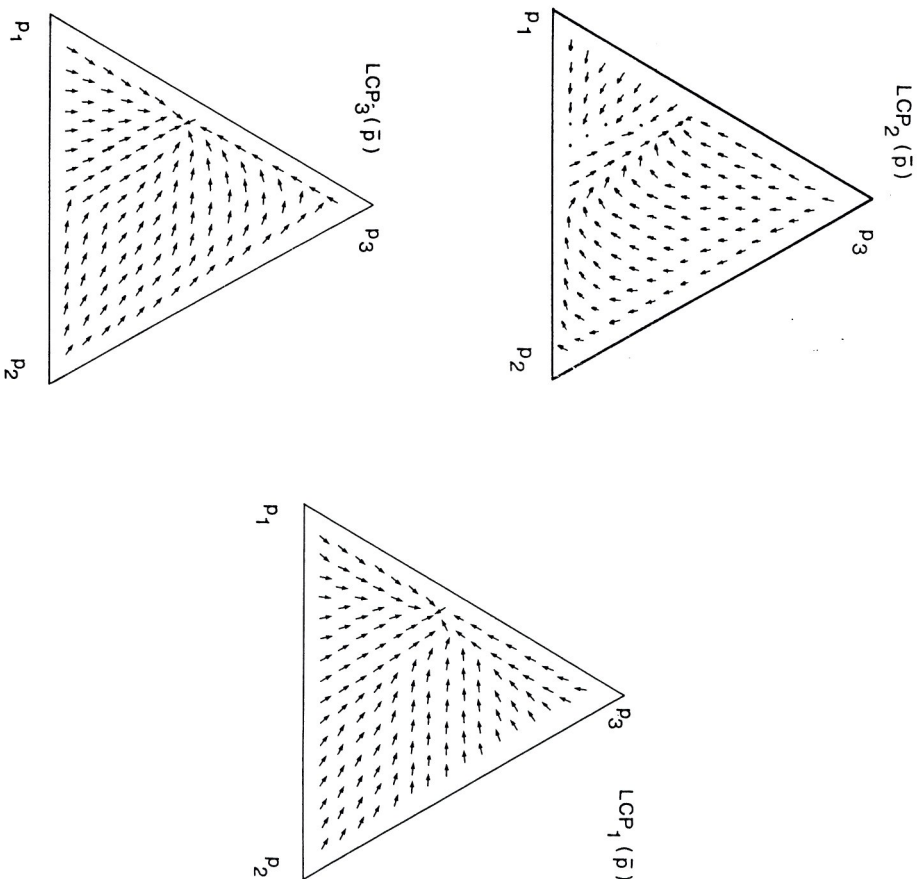


Fig. 2. The vector-fields of $LCP_i(\bar{p})$, $i = 1, 2, 3$.

Even for this very simple example, the analytic solutions get messy. Let us therefore switch to a computational presentation. Consider a fine grid on the interior of the price simplex. Construct and solve $LCP_i(\bar{p})$ at each such grid point, $\bar{p}$, and mark the direction, $(\bar{p} - \bar{p})/\|\bar{p} - \bar{p}\|$, to the computed solution by an arrow based at $\bar{p}$. We will denote the resulting map a "vector-field". Termination with a secondary ray is indicated by a dot, i.e., the lack of an arrow.

Figure 2 displays the vector-fields for the three alternative processes. It confirms our findings for $LCP_2(\bar{p})$ and conveys much better than algebra the peculiar convergence properties of the algorithm applied to this problem. In all cases the parameters $b = (0, 5, 3)^T$ and $\alpha = 0.9$ are used.

Again there are several observations to make. First, except for the RAY-region for $LCP_2(\bar{p})$, the vector-fields are continuous and bend smoothly, although rather sharply in a few places. The global convergence is evident, for both $LCP_1(\bar{p})$ and $LCP_3(\bar{p})$, as is the convergence for $LCP_2(\bar{p})$ when initiated outside the RAY-region.

Second, we observe that the direction taken at a given point $\bar{p}$ may be quite different from one LCP (or numéraire) to another. It turns out that the computational effort to obtain the equilibrium varies considerably between the numéraires. In some parts of the simplices, arrows are nearly orthogonal to the equilibrium direction. The next iterate may be further from the equilibrium, and convergence is not monotone.¹³

Third, the RAY-region for $LCP_2(\bar{p})$ is a zone where the directions of the vector-fields on either side are opposed to each other. In CS(2) (to the left) the arrows point downwards, while in CS(3) they point upwards. No other area of the three simplices has this feature.

Let us explore this latter feature by reducing the budget share α . As a function of α , the equilibrium moves downward along the line $p_1 = p_2 + p_3$, implied by $y = 3 > 0$.¹⁴ Figure 3 shows the vector-field for $LCP_2(\bar{p})$ for $\alpha = 0.7$. As mentioned, there is no RAY-region for $\alpha \leq \frac{2}{3}$. The right part of the figure is a "magnified view" of the region of interest and shows that the vector-field is continuous and bends smoothly. Figure 4 displays the analogous results for $LCP_3(\bar{p})$ and exhibits the same features.

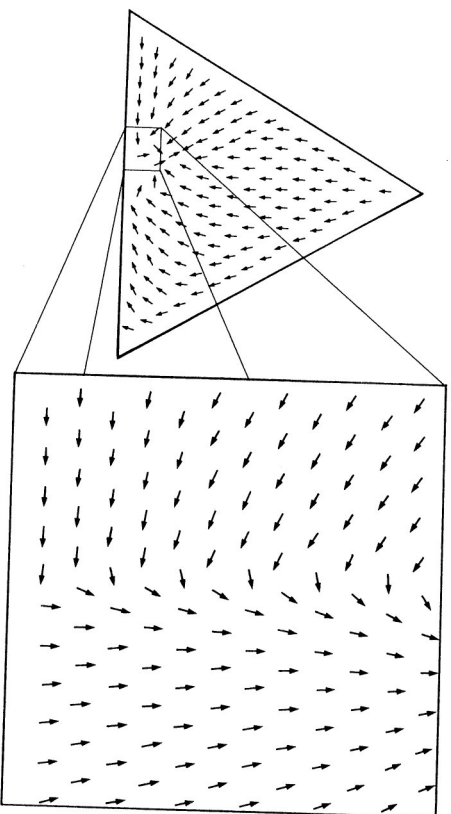


Fig. 3. The vector-field of $LCP_2(\bar{p})$ for $\alpha = 0.7$.

Reducing α to 0.61, $LCP_3(\bar{p})$ has a secondary ray. Figure 5 shows the whole simplex and focuses on the "trouble" region. It indicates that this RAY-region is a tiny band between two regions with conflicting directions.

Let us end this section with some examples of a whole sequence of iterates of the algorithm. Figure 6 shows the iterates from different starting points when using $LCP_2(\bar{p})$. Figure 7 displays the sequences from a given starting point, but for the

¹³ Take $\bar{p} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^T$ in $LCP_2(\bar{p})$. The scaled solution is $\bar{p} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})^T$ which is further from the equilibrium according to as well the L_1 and L_2 as L_∞ norm.

¹⁴ For $\alpha = 1$, $p^* = (\frac{1}{3}, 0, \frac{2}{3})^T$ and for $\alpha = 0.6$, $p^* = (\frac{1}{3}, \frac{1}{3}, 0)^T$.

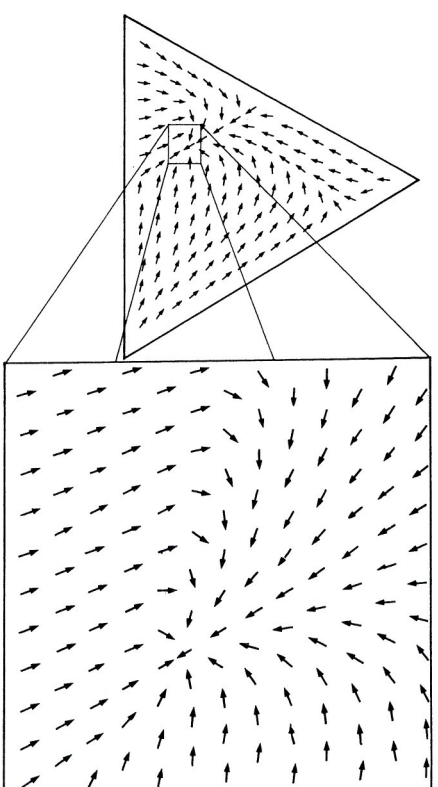


Fig. 4. The vector-field of $LCP_3(\bar{p})$ for $\alpha = 0.7$.

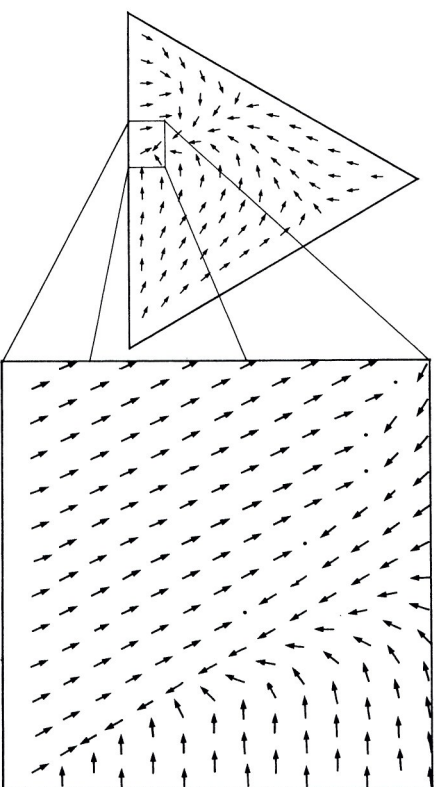


Fig. 5. The vector-field of $LCP_3(\bar{p})$ for $\alpha = 0.61$.

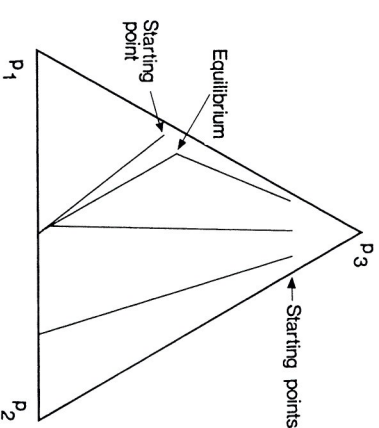


Fig. 6. Sequences of iterates for $LCP_2(\bar{p})$.

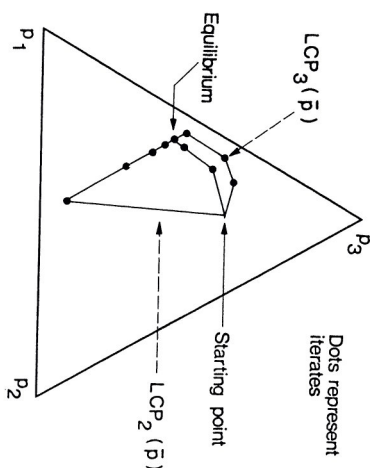


Fig. 7. Sequences of iterates for $LCP_i(\bar{p})$, $i = 1, 2, 3$.

3 different commodities being numéraire. Again we have chosen the parameter values $b = (0, 5, 3)^T$ and $\alpha = 0.9$.

6. Conclusion

We have considered the application of a sequential LCP algorithm to a constructed Walrasian equilibrium problem. By a combination of algebra and numerical methods we have shown that when applied to this problem, the algorithm converges globally to the equilibrium for certain values of the parameters.

We demonstrated that some LCPs may have no solution, and that other LCPs have both a complementary solution and a homogeneous solution. We observed in the latter cases that a complementary solution is computed. Only when there is no complementary solution, does Lemke's algorithm terminate with a secondary ray.

It seems essential that both q and M of the LCP are considered for characterizing the performance of Lemke's algorithm to these LCPs. Even if M admits homogeneous solutions, q may rule out the existence of a secondary ray. A fundamental observation is that

For any $\bar{p} \in \bar{S} = \{p \mid \sum_{i=1}^m p_i = 1, p_i > 0\}$, there are m alternative LCPs denoted $LCP_i(\bar{p})$, $i = 1, \dots, m$, from which a subsequent iterate may possibly be computed.

Thus, if a particular $LCP_i(\bar{p})$ at some point $\bar{p}$ has no solution, there are alternative LCPs at this point, and (in this example) at least one of these has a solution. Hence a SLCP process that switches numéraire and solves another LCP whenever a secondary ray is encountered will succeed.

We stated four propositions of varying generality. The first is particular to our example problem. The next two propositions are valid for the example for any b_2 , $b_3 > 0$ and $0 < \alpha < 1$.

Lemke's algorithm computes a solution to $LCP_i(\bar{p})$, $i = 1, 2, 3$, if there is one.

For any $\bar{p} \in \bar{S}$, there is an $LCP_i(\bar{p})$ (for some i), that has a solution.

The generalization of the first of these to other models is confirmed by all our experiments reported in [7, 8, 9].

The fourth proposition holds for the example of Section 3 with $b = (0, 5, 3)^T$ and $0 < \alpha < 1$, and appears to hold for other values as well.

For any $\bar{p} \in \bar{S}$, the SLCP algorithm that switches numéraire if necessary, converges to the equilibrium.

There is, however, one counterexample to the latter two propositions. For some parameter values of Scarf's unstable exchange equilibrium model [14], there are points $\bar{p}$ for which no $LCP_i(\bar{p})$ has a solution, [8].¹⁵ Next, for other points $\bar{p}$ the LCP has two solutions and Lemke's algorithm computes the "wrong" one.¹⁶ This is the trivial solution, i.e., $w = q$, $z = 0$, which of course cannot be the equilibrium. A sequence of (modified) iterates based on this solution cannot converge to the equilibrium. These examples indicate that Lemke's algorithm may not compute the "best" solution in the case where the LCP has multiple solutions (while the equilibrium is unique). Hence the convergence of SLCP may be jeopardized.¹⁷

In view of the demonstrated efficiency of the SLCP-process, we will conclude that provision of additional insight into theoretical aspects of this algorithm seems worthwhile. For example, under what conditions on utility and production functions do the last two propositions hold true?

Acknowledgement

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¹⁵ Solutions to the equations ($w = q + Mz$) are outside the positive orthant.

¹⁶ Observe in Figure 1 that, for $\bar{p} \in CS(2)$ (lower left), Lemke's algorithm computes the solution $\bar{p} = (\frac{1}{2}, \frac{1}{2}, 0)^T$, although these LCPs have another solution that is closer to the equilibrium.

¹⁷ Scarf's model is designed as a counter-example to stability of the tâtonnement process. For a wide range of parameter values, the SLCP process converges globally, but not for all parameters. Experiments with Brodie's OCTASOLV fixed-point code indicate that even such algorithms may fail. That is, the values of excess demands are arbitrarily low, but still nonzero at nonequilibrium points. Such ill-conditioned examples make it difficult to distinguish between failures due to the concept of an algorithm and failures due to computational inaccuracies.

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